DATA ANALYTICS LAB SHEET 3

Name: ANURAG PARMAR

Roll No.: 8

Section: B

CU ID: CU24250076

B.Tech CSE

Experiment No. 3

Statistical Analysis and Hypothesis Testing using Python

AIM

To perform statistical analysis and hypothesis testing on a real-world dataset using Python in order to identify relationships between variables, validate assumptions, and support data-driven decision-making.

DATASET

Titanic Survival Dataset.

PURPOSE

Statistical analysis is used to summarize data, identify relationships, test hypotheses, and determine whether observed patterns are statistically significant or may have occurred by chance.

1. Import Required Python Libraries

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm

print("Libraries imported successfully.")
```

2. Load the Real-World Dataset

Code:

```
url = "https://raw.githubusercontent.com/mwaskom/seaborn-data/master/titanic.csv"

df = pd.read_csv(url)

print("Dataset loaded successfully.")
print("Dataset shape:", df.shape)
display(df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

3. Calculate Descriptive Statistics

Code:

```
data = df[["survived", "age", "fare", "pclass"]].dropna()

print("Mean:")
display(data[["age", "fare"]].mean())

print("Median:")
display(data[["age", "fare"]].median())

print("Mode:")
display(data[["age", "fare"]].mode().iloc[0])

print("Variance:")
display(data[["age", "fare"]].var())

print("Standard Deviation:")
display(data[["age", "fare"]].std())

print("Complete descriptive statistics:")
display(data[["age", "fare"]].describe())
```

```
age     29.699118
fare    34.694514
dtype: float64
```

```
age     28.0000
fare    15.7417
dtype: float64
```

```
age     24.0
fare    13.0
Name: 0, dtype: float64
```

```
age      211.019125
fare    2800.413100
dtype: float64
```

```
age     14.526497
fare    52.918930
dtype: float64
```

```
              age        fare
count  714.000000  714.000000
mean    29.699118   34.694514
std     14.526497   52.918930
min      0.420000    0.000000
25%     20.125000    8.050000
50%     28.000000   15.741700
75%     38.000000   33.375000
max     80.000000  512.329200
```

4. Pearson Correlation Analysis

Code:

```
r, p_corr = stats.pearsonr(data["age"], data["fare"])

print("Pearson correlation coefficient:", round(r, 4))
print("Correlation p-value:", round(p_corr, 6))

correlation_matrix = data[["age", "fare", "pclass"]].corr()
display(correlation_matrix)
```

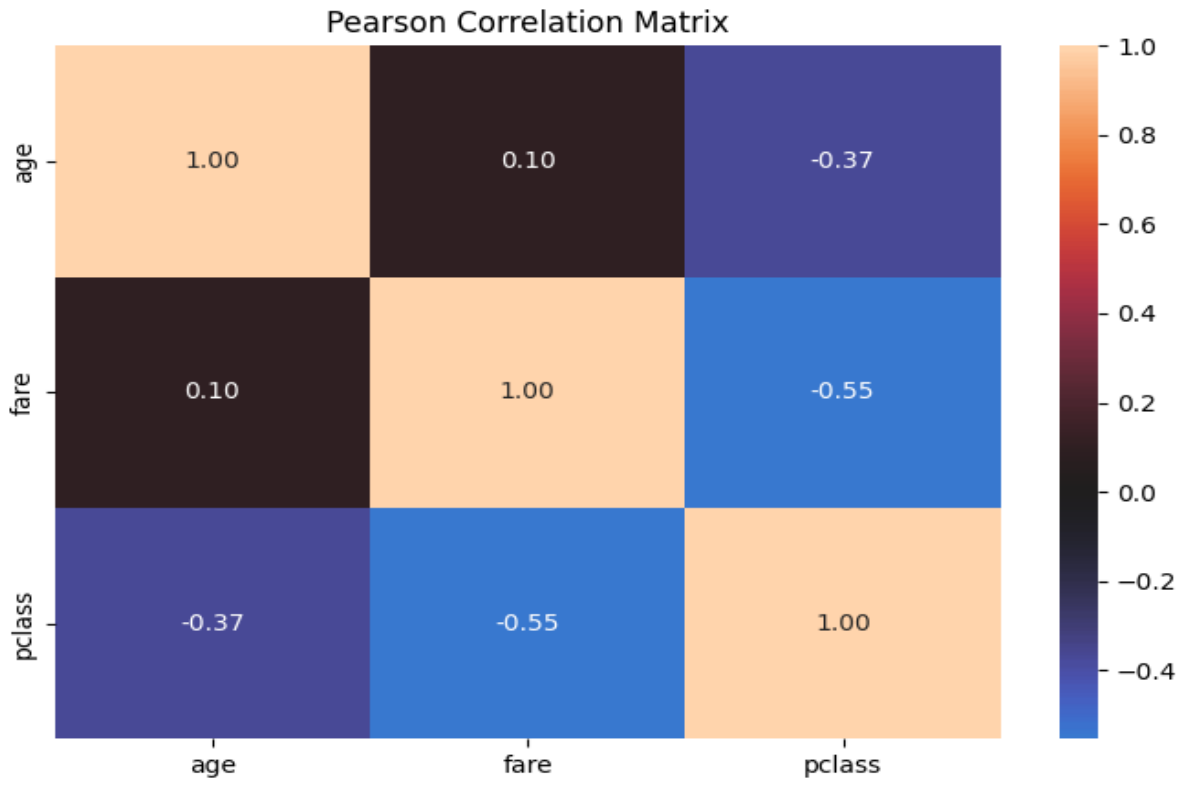
```
             age      fare    pclass
age     1.000000  0.096067 -0.369226
fare    0.096067  1.000000 -0.554182
pclass -0.369226 -0.554182  1.000000
```

5. Visualize the Correlation Matrix

Code:

```
plt.figure(figsize=(7, 5))
sns.heatmap(
    correlation_matrix,
    annot=True,
    fmt=".2f",
    center=0
)
plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()
```

```
<Figure size 700x500 with 2 Axes>
```



6. Formulate Null and Alternative Hypotheses

Business question:

Is there a significant difference in average fare between passengers who survived and passengers who did not survive?

Null Hypothesis (H■): The mean fare of survivors and non-survivors is equal.

Alternative Hypothesis (H■): The mean fare of survivors and non-survivors is different.

Significance level (α): 0.05

Code:

```
survived_fare = data.loc[data["survived"] == 1, "fare"]
not_survived_fare = data.loc[data["survived"] == 0, "fare"]

print("Mean fare of survivors:", round(survived_fare.mean(), 2))
print("Mean fare of non-survivors:", round(not_survived_fare.mean(), 2))
```

7. Independent Sample t-Test

Code:

```
t_stat, p_value = stats.ttest_ind(
    survived_fare,
    not_survived_fare,
    equal_var=False
)

alpha = 0.05

print("t-statistic:", round(t_stat, 4))
print("p-value:", round(p_value, 6))

if p_value < alpha:
    print("Decision: Reject H0.")
    print("Conclusion: There is a statistically significant difference in mean fare between the two groups.")
else:
    print("Decision: Fail to reject H0.")
    print("Conclusion: There is not enough evidence of a statistically significant difference in mean fare between the two grou
```

8. One-Way ANOVA

Code:

```
class1 = data.loc[data["pclass"] == 1, "fare"]
class2 = data.loc[data["pclass"] == 2, "fare"]
class3 = data.loc[data["pclass"] == 3, "fare"]

f_stat, p_anova = stats.f_oneway(class1, class2, class3)

print("F-statistic:", round(f_stat, 4))
print("p-value:", round(p_anova, 6))

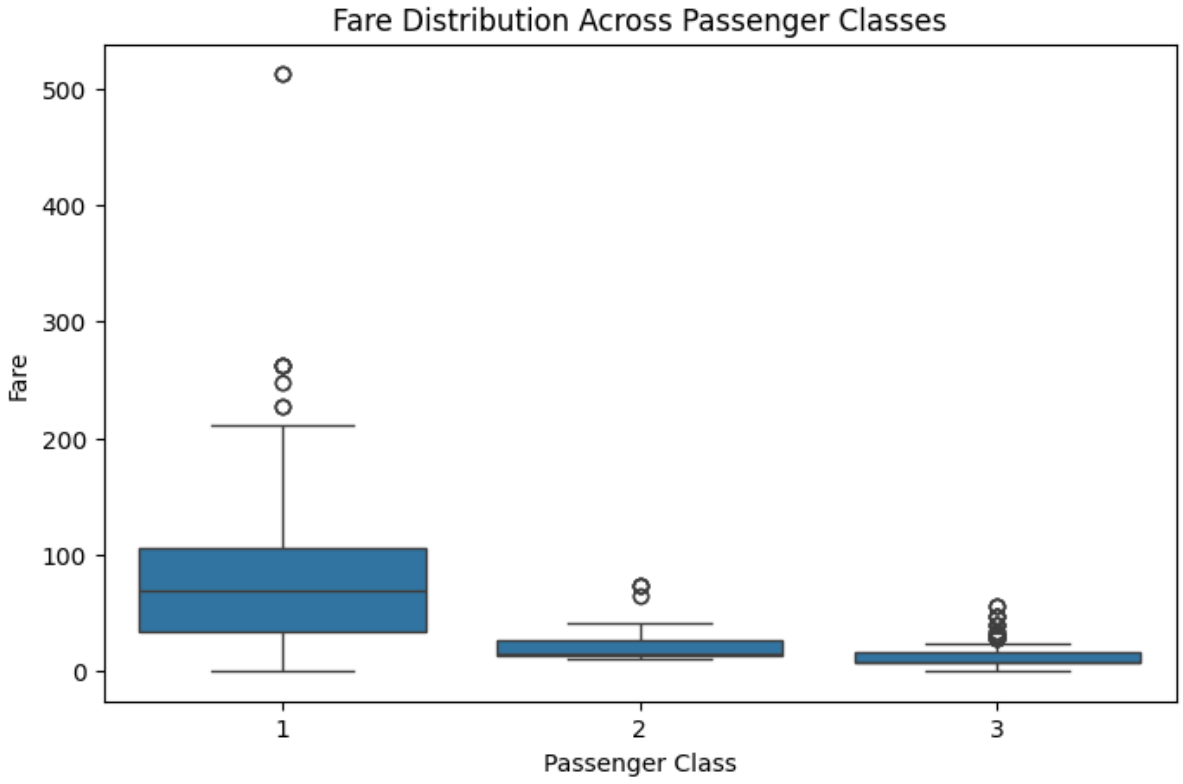
if p_anova < alpha:
    print("Decision: Reject H0.")
    print("Conclusion: At least one passenger class has a statistically different mean fare.")
else:
    print("Decision: Fail to reject H0.")
    print("Conclusion: There is not enough evidence of a statistically significant difference among the class means.")
```

9. Visualize Group Differences

Code:

```
plt.figure(figsize=(8, 5))
sns.boxplot(data=data, x="pclass", y="fare")
plt.title("Fare Distribution Across Passenger Classes")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")
plt.show()
```

```
<Figure size 800x500 with 1 Axes>
```



10. Build a Simple Linear Regression Model

Code:

```
# Analyze the relationship between Age and Fare
X = sm.add_constant(data["age"])
y = data["fare"]

model = sm.OLS(y, X).fit()

print(model.summary())
```

11. Regression Coefficients, Confidence Intervals and R^2

Code:

```
print("Intercept:", round(model.params["const"], 4))
print("Age coefficient:", round(model.params["age"], 4))
print("R²:", round(model.rsquared, 4))
print("Adjusted R²:", round(model.rsquared_adj, 4))
```

```
print("\n95% Confidence Intervals:")
display(model.conf_int())
```

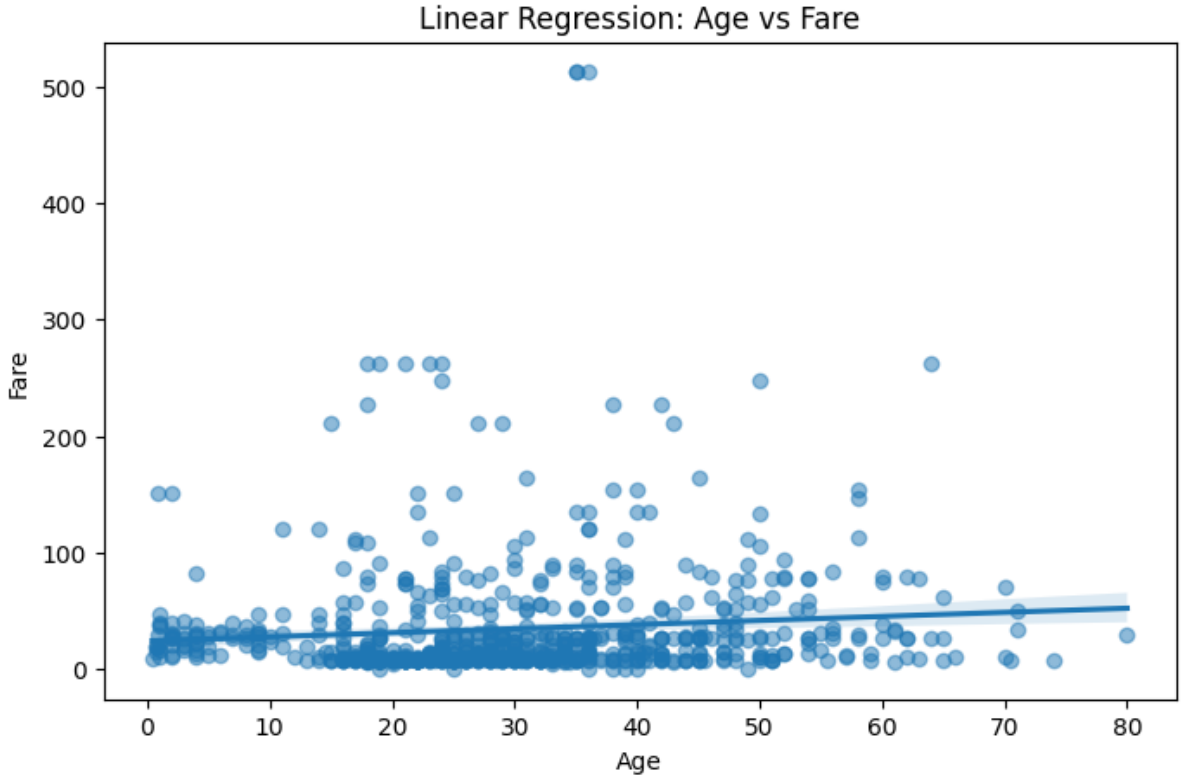
```
               0          1
const  15.481351  33.120452
age     0.083165   0.616762
```

12. Regression Visualization

Code:

```
plt.figure(figsize=(8, 5))
sns.regplot(
    data=data,
    x="age",
    y="fare",
    scatter_kws={"alpha": 0.5}
)
plt.title("Linear Regression: Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

```
<Figure size 800x500 with 1 Axes>
```



13. Summary of Statistical Results

Code:

```
print("STATISTICAL ANALYSIS SUMMARY")
print("-------------------------------")
print("Pearson correlation coefficient:", round(r, 4))
print("Pearson correlation p-value:", round(p_corr, 6))
print("t-test p-value:", round(p_value, 6))
print("ANOVA p-value:", round(p_anova, 6))
print("Regression R²:", round(model.rsquared, 4))

print("\nInterpretation:")
print("A p-value below 0.05 is considered statistically significant at the 5% significance level.")
print("R² represents the proportion of variation in Fare explained by Age in this simple regression model.")
```

Answers to Questions

Q1. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize and describe the characteristics of observed data using measures such as mean, median, mode, variance, and standard deviation. Inferential statistics use sample data to draw conclusions about a larger population using methods such as hypothesis testing and confidence intervals.

Q2. Explain the concepts of the Null Hypothesis (H■) and Alternative Hypothesis (H■).

The Null Hypothesis (H■) is the default statement that there is no significant difference, relationship, or effect. The Alternative Hypothesis (H■) states that a significant difference, relationship, or effect exists.

Q3. What is a p-value? How is it used to make statistical decisions?

A p-value is the probability of obtaining a result at least as extreme as the observed result when the null hypothesis is true. At a chosen significance level such as 0.05, if the p-value is less than 0.05, H■ is rejected. Otherwise, H■ is not rejected.

Q4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

A t-test is generally used to compare the means of two groups. ANOVA is used to compare the means of three or more groups. For example, a t-test can compare the average income of two groups, while ANOVA can compare average income across several departments.

Q5. What does the Pearson correlation coefficient indicate? What are its possible values?

The Pearson correlation coefficient measures the direction and strength of a linear relationship between two numerical variables. Its values range from -1 to +1. A value near +1 indicates a strong positive relationship, a value near -1 indicates a strong negative relationship, and a value near 0 indicates little or no linear relationship.

Q6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

R^2 indicates the proportion of variation in the dependent variable that is explained by the independent variable or variables in the regression model. Its value generally ranges from 0 to 1 for standard regression models. A higher R^2 means that a larger proportion of variation is explained by the model.

Q7. Why is statistical analysis important before applying machine learning algorithms?

Statistical analysis helps understand the data, identify relationships, detect unusual observations, check assumptions, and select useful features. It can also reveal data-quality issues and guide the choice of preprocessing and modeling techniques.

Q8. What assumptions should be satisfied before performing a t-test or ANOVA?

Important assumptions include independence of observations, appropriate measurement of the dependent variable, approximate normality within groups, and suitable variance assumptions. For standard ANOVA, homogeneity of variances across groups is also commonly considered.

Q9. How can regression analysis help organizations in forecasting and decision-making?

Regression analysis can quantify relationships between variables and estimate how changes in predictor variables are associated with changes in an outcome. Organizations can use these relationships for forecasting, planning, budgeting, pricing, resource allocation, and other data-driven decisions.

Q10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

1. A company can test whether a new marketing campaign produces a significant change in sales compared with the previous campaign.

2. A business can test whether average customer satisfaction differs significantly between two products, services, or customer groups.